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$$\begin{aligned} I &= \int \frac{x}{2x^2 - 3x + 1} dx \\ &= \int \frac{x}{(2x - 1)(x - 1)} dx \end{aligned}$$

We gebruiken de regel van Fuss:

$$\begin{aligned} A &= \left(\frac{x}{x - 1} \right) \Big|_{x=\frac{1}{2}} = -1 \\ B &= \left(\frac{x}{2x - 1} \right) \Big|_{x=1} = 1 \\ &\Rightarrow \frac{x}{(2x - 1)(x - 1)} = \frac{-1}{2x - 1} + \frac{1}{x - 1} \\ I &= \int \left(\frac{1}{x - 1} - \frac{1}{2x - 1} \right) dx \\ t &= 2x - 1 \Rightarrow \frac{1}{2} dt = dx \\ I &= \ln |x - 1| - \frac{1}{2} \int \frac{1}{t} dt \\ &= \ln |x - 1| - \frac{1}{2} \ln |2x - 1| + c \end{aligned}$$

References